EXPERIMENT 7

IMPLEMENTATION OF HYBRID TOPOLOGY (BUS AND RING) USING PACKET TRACER

Aim

To implement a hybrid topology (bus and ring) using Packet Tracer and to transmit data between the devices connected in the hybrid topology.

Software Used

Cisco Packet Tracer (end devices, hubs, switches, connecting cables)

Procedure

Step 1: Start Cisco Packet Tracer.

Step 2: Click **End Devices**, select the Generic PC and place four PCs in the workspace.

Step 3: Building the bus part: click **Hubs**, place two hubs (Hub0 and Hub1) and connect them to each other. Connect PC0 and PC1 to Hub0.

Step 4: Building the ring part: click **Switches**, place four switches (Switch0 to Switch3) and connect them one after another in a closed loop. Connect PC2 to one switch and PC3 to another switch of the ring.

Step 5: Connect Hub1 to Switch0 so that the bus part and the ring part are joined to form the hybrid topology. Use the Copper Straight-Through cable for all connections.

Step 6: Click each PC, open the **Config** tab, select FastEthernet0 and type the IP address. The subnet mask is generated automatically.

Step 7: In **Realtime** mode, select the **Add Simple PDU** tool and click PC0 and then PC3. The PDU Last Status should show **Successful**.

Step 8: In **Simulation** mode, keep only the ICMP filter, send the PDU from PC0 to PC3 and click Capture/Forward until the ping is completed. The ICMP packet moves through the hubs and switches and the status shows **Successful**.

Execution Screenshot

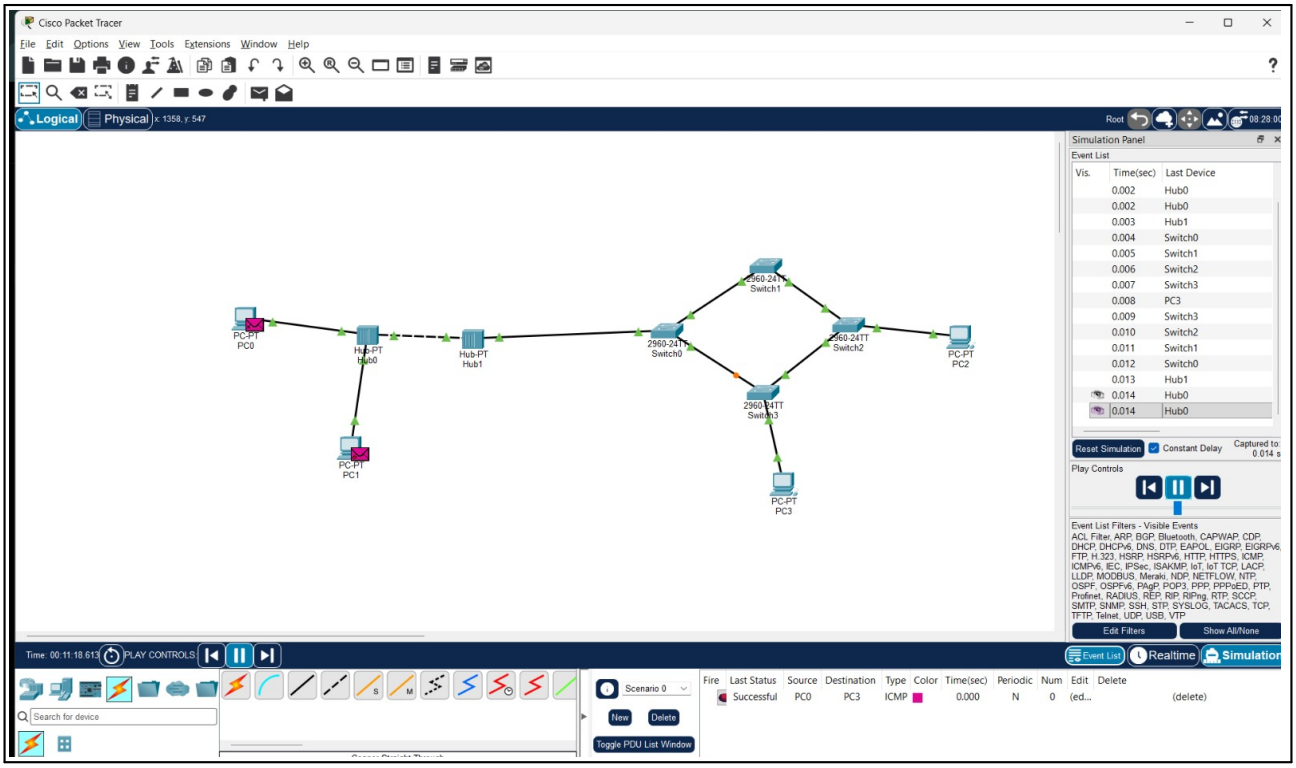


Fig: Hybrid topology (bus and ring) with ICMP ping from PC0 to PC3

Result

Thus the hybrid topology is implemented with the Packet Tracer simulation tool.